

Yash Swaminathan

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Education

University of Waterloo

Bachelor of Applied Science (BASc) in Systems Design Engineering (Co-op)

Waterloo, ON

Sep. 2024 – Present

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Go, Java, C++, HTML/CSS

Frameworks & Databases: FastAPI, Node.js, Express.js, PostgreSQL, Redis, Prisma, Spring Boot, React, Next.js, Socket.io

Infrastructure & Tools: Docker, AWS, Terraform, Github Actions, Prometheus, Grafana, New Relic, k6, MinIO, Git, Linux

Libraries: Stripe SDK, NumPy, Pandas, TensorFlow, Pytest, Jest, Pydantic, PyYAML, jsonschema

Experience

Software Engineer Intern — Backend Integrations

StackAdapt

Incoming May 2026

Toronto, ON

Software Engineer Intern — Backend & DevOps

Micromart

Aug 2025 – Dec 2025

Toronto, ON

- Modelled infrastructure capacity for **10,000 machines** through **performance analysis and cost modelling**, enabling fleet expansion from **900 → 2,100 machines in one month** and deployment of **\$7.2M in hardware**.
- Designed **k6 load tests** on TypeScript/Node.js backend simulating **20,000 RPS** and **15,000 concurrent WebSocket connections**, targeting API calls consuming **95% of total backend request time** to identify scalability bottlenecks.
- Built observability pipelines in **New Relic and Grafana** with distributed tracing and custom dashboards tracking **API latency, resource utilization, cron jobs, and service health**, reducing root cause identification time by **60%**.
- Optimized backend performance through **Redis caching**, refactoring system heartbeat endpoints, and **retry/backoff logic** on machine restart/shutdown, reducing **P95 latency by 35%** and total **API call time by 15%**.

Software Developer Intern — Backend & DevOps

Turing

Jan 2025 – Apr 2025

Calgary, AB

- Designed and implemented **Alembic Migration** scripts for **PostgreSQL** test databases, automating test database creation and configuration, reducing manual setup time by **200+** hours, resulting in **\$15,000+** of savings.
- Developed **10+ RESTful API endpoints (FastAPI)** and authored **60+** unit tests to achieve **95%** coverage, ensuring reliable API behaviour and maintenance of critical features.
- Resolved **20+** technical issues (Docker misconfigs, API bottlenecks, flaky tests), improving system stability and reliability.

Data Science Intern

Gradiant (Synauta)

Sep 2023 – Jan 2024

Calgary, AB

- Developed a **Python-based analytics pipeline** to process **IoT sensor data**, identifying optimal cartridge filter replacement timing and reducing premature replacements by **15–20%**.
- Designed and deployed a **Dockerized pipeline (Pandas, NumPy, SciPy)** leveraging **cron jobs** to automate replacement scheduling and eliminate **90%** of manual tracking processes.

Projects

🔗 **ChatterBox - Real-Time Messaging Platform** | *Node.js, Express, PostgreSQL, Redis, Socket.io, Docker, MinIO, Jest, k6*

- Built real-time messaging backend with **Redis pub/sub** enabling horizontal scaling across multiple servers, supporting **1000+ concurrent WebSocket connections** through efficient room-based message broadcasting.
- Engineered **group messaging system** with admin roles, participant management, and PostgreSQL full-text search with @mention parsing, using advisory locks to prevent race conditions in groups with **100+** members.
- Implemented **JWT-based authentication** with **bcrypt password hashing**, refresh token rotation, and device-based session management for secure multi-device access.

🔗 **E-Commerce Platform** | *Java, Spring Boot, Go, TypeScript, React (Next.js), Express.js, Prisma, Stripe SDK*

- Designed event-driven microservices architecture (**Go + Spring Boot**) that reduced order-processing latency from **800ms to 480ms**, enabling real-time inventory updates and preventing overselling across **10+ concurrent orders**.
- Implemented secure payment flow with **Stripe webhooks** for async order confirmation and **AWS S3** for product media, processing **1,000+ test transactions** with automated payment reconciliation.
- Automated deployment pipeline with **GitHub Actions** and **Terraform**, achieving **60% faster releases** (25min → 10min) with zero-downtime blue-green deployments and automated rollback capabilities.

🔗 **File Management & Validation System** | *Python, FastAPI, PostgreSQL, React, Docker, Poetry, HTML/CSS*

- Built schema comparison engine using **DeepDiff** and **PyYAML** to detect configuration drift between environments, generating diff reports in **sub-200ms** for nested YAML structures stored in **PostgreSQL**.
- Implemented validation pipeline with **jsonschema** and **Pydantic**, enforcing type safety through custom field validators.